
Linear Regression using DAX

In the lessons in Week 4, we saw how to find the values β_1 and β_0 for the regression line using the advertising example.

The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS. Using some calculus, one can show that the minimizers are

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x},\end{aligned}\tag{3.4}$$

where $\bar{y} \equiv \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$ are the sample means. In other words, (3.4) defines the *least squares coefficient estimates* for simple linear regression.

The final equation representing the link between TV and Sales was

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}.$$

Where β_0 was 0.06 and β_1 was about 7.

In this task, you will have to repeat my steps to work out the β_0 and β_1 values for Radio and Newspaper Media.

Then compare the three β_1 to check to which media corresponds a “faster growth” (steeper gradient).

Use the link below if you need to watch the demo again

[Link to the Demo done in the lesson](#)

I would suggest you open two different Power BI projects (one for Newspaper and one for Radio) so that you can use the same terminology I used in the lesson without conflicts (otherwise you will have two columns named x , one for Newspaper and one for Radio, and two named y , which will generate an error – you can’t have two columns with the same name in the same Project).

Please note that this task is quite advanced and if you haven’t done much Maths in the past you might find it a bit challenging. Please embrace the challenge and use it as an opportunity to learn something new. :)

Once you have the solution, take a screenshot and post it on Padlet. You should answer the question below:

Which media provides the fastest growth in Sales? – Compare your β_1 values to answer it. Good Luck!